

CTD QUIZ-3 SOLUTION (SET-A)

Sol: Given that—

The balanced Δ -connected load demanding total apparent power,

$$|S| = 10 \text{ kVA}$$

→ (1 Mark)

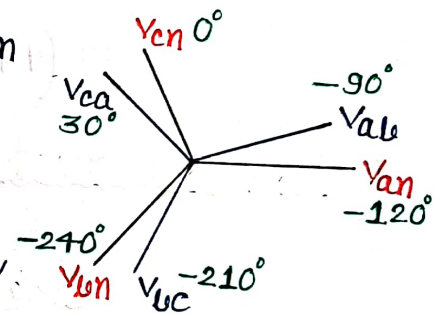
$$\text{Line voltage, } V_{ca} = 160 \angle 30^\circ \text{ V}$$

$$\text{PF} = 0.91 \text{ (lagging)}$$

$$\cos(\theta - \phi) = 0.91$$

$$\therefore (\theta - \phi) = \cos^{-1}(0.91) = 24.5^\circ \rightarrow (1 \text{ Mark})$$

Positive phase sequence; V_{an}, V_{bn}, V_{cn}



Hence, Line voltage, $V_L = |V_{ca}| = 160 \text{ V}$

$$|V_{an}| = \frac{V_L}{\sqrt{3}} = \frac{160}{\sqrt{3}} = 92.38 \text{ V}$$

→ (1 Mark)

According to positive phase sequence,

$$V_{cn} = 92.38 \angle 0^\circ \text{ V}$$

$$V_{an} = 92.38 \angle -120^\circ \text{ V}$$

→ (2 Mark)

$$V_{bn} = 92.38 \angle -240^\circ \text{ V}$$

As we know — $|S| = \frac{\sqrt{3}}{2} |V_L| \cdot |I_L|$

$$\therefore |I_L| = \frac{2|S|}{\sqrt{3}|V_L|} = \frac{2 \times 10 \times 10^3}{\sqrt{3} \times 160} = 72.17 \text{ A}$$

$$\therefore I_L = |I_{aA}| = |I_{bB}| = |I_{cC}| = 72.17 \text{ A}$$

→ (1 Mark)



As we know — $I_p = \frac{I_L}{\sqrt{3}}$ (for Δ -connected load)

$$I_p = \frac{72.17}{\sqrt{3}} = 41.66 \text{ A}$$

$\therefore |I_{CA}| = |I_{AB}| = |I_{BC}| = I_p = 41.66 \text{ A} \rightarrow (1 \text{ Mark})$
According to positive phase sequence,

$$I_{CA} = V_{CA}/Z_p, \angle I_{CA} = \angle V_{CA} - \angle Z_p = \angle 30^\circ - \angle 24.5^\circ$$

$$I_{CA} = 41.66 \angle 30^\circ - 24.5^\circ = 41.66 \angle 5.5^\circ \text{ A}$$

$$I_{AB} = 41.66 \angle -90^\circ - 24.5^\circ = 41.66 \angle -114.5^\circ \text{ A} \rightarrow (1 \text{ Mark})$$

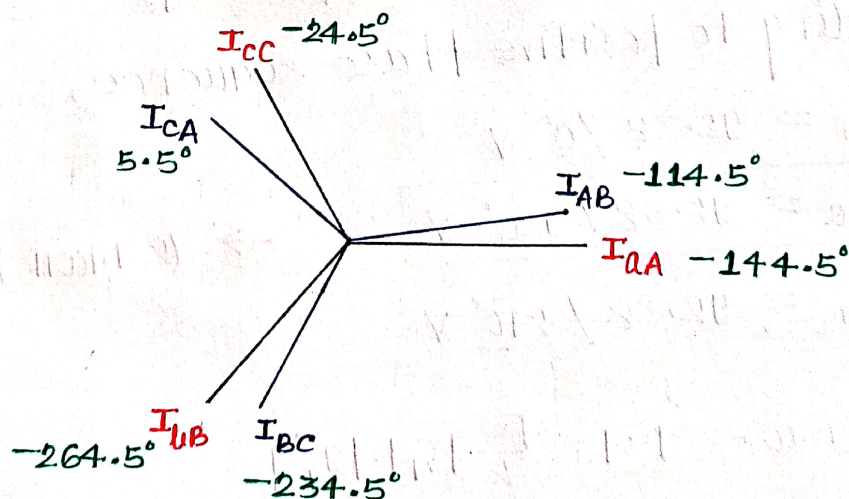
$$I_{BC} = 41.66 \angle -210^\circ - 24.5^\circ = 41.66 \angle -234.5^\circ \text{ A} \rightarrow (1 \text{ Mark})$$

$$\therefore I_{bB} = I_{BC} - I_{AB}$$

$$= (41.66 \angle -234.5^\circ) - (41.66 \angle -114.5^\circ)$$

$$= 72.16 \angle 95.5^\circ \text{ A}$$

$$= 72.16 \angle -264.5^\circ \text{ A} \rightarrow (1 \text{ Mark})$$



CTD QUIZ-3 SOLUTION (SET-B)

Sol÷ Given that -

$$S_L = S_{\text{load}} = (1800 + j700) \text{ W}$$

$$S_S = S_{\text{source}} = (1850 + j700) \text{ W}$$

$$R_W = 1.2 \Omega$$

Positive phase sequence; V_{an}, V_{bn}, V_{cn}

$$\therefore |S_L| = \sqrt{1800^2 + 700^2} = 1931.32 \text{ VA}$$

$$1800 = |S_L| \cdot \cos(\theta - \phi) = P_{\text{avg}_L} = \text{Re}\{S\}$$

$$\text{PF} = \cos(\theta - \phi) = 0.93$$

$$\therefore (\theta - \phi) = \cos^{-1}(0.93) = 21.25^\circ \rightarrow (0.5 \text{ MARK})$$

$$\begin{aligned} \text{Power loss due to lines, } P_{\text{loss}} &= P_S - P_L \\ &= 1850 - 1800 = 50 \text{ W} \end{aligned}$$

$$P_{\text{loss}} = 3(I_L^2 \cdot R_W) \rightarrow (0.5 \text{ MARK})$$

$$50 = 3(I_L^2 \times 1.2)$$

$$\therefore I_L = 3.73 \text{ A rms} \rightarrow (0.5 \text{ MARK})$$

$$\text{As we know - } I_p = \frac{I_L}{\sqrt{3}} = 2.15 \text{ A rms} \rightarrow (1 \text{ MARK})$$

$$\text{As we know - } P_S = \sqrt{3} V_L I_L \cos(\theta - \phi) \rightarrow (0.5 \text{ MARK})$$

$$1850 = \sqrt{3} \times V_L \times 3.73 \times 0.93$$

$$V_L = 304.63 \text{ V rms} \rightarrow (1.5 \text{ MARK})$$

According to positive phase sequence,

$$V_{an} = \left(\frac{V_L}{\sqrt{3}}\right) \angle 0^\circ \text{ V} \rightarrow (0.5 \text{ Mark})$$

$$\therefore V_{an} = 175.87 \angle 0^\circ \text{ V rms} \rightarrow (1 \text{ Mark})$$

$$\therefore V_{AB} = V_{ab} = \sqrt{3} \cdot V_{an} \angle 30^\circ = 304.62 \angle 30^\circ \text{ V rms} \rightarrow (0.75 \text{ Mark})$$

$$\therefore Z_p = \frac{V_p}{I_p} \angle 21.25^\circ = \frac{|V_{AB}|}{I_p} \angle 21.25^\circ$$
$$= \frac{304.62 \angle 21.25^\circ}{2.15}$$

$$= 141.68 \angle 21.25^\circ \Omega \rightarrow (0.75 \text{ Mark})$$

$$\therefore I_{AB} = \frac{V_{AB}}{Z_p} = \frac{304.62 \angle 30^\circ}{141.68 \angle 21.25^\circ}$$
$$= 2.15 \angle 8.75^\circ \text{ A rms} \rightarrow (0.5 \text{ Mark})$$

As we know —

$$I_{aA} = \sqrt{3} I_{AB} \angle -30^\circ \rightarrow (0.5 \text{ Mark})$$

$$= \sqrt{3} \times 2.15 \angle 8.75^\circ \times \angle -30^\circ$$

$$= 3.73 \angle -21.25^\circ \text{ A rms} \rightarrow (1 \text{ Mark})$$

